



Tachycardia

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Tachycardia Definition & Types

Definition

- Resting heart rate >100 bpm in adults
- May be physiological (exercise, fever, anxiety) or **pathological**

• Major Types

- **Sinus Tachycardia** – Normal rhythm, elevated rate
- **Supraventricular Tachycardia (SVT)** – AVNRT, AVRT, atrial tachycardia
- **Atrial Fibrillation/Flutter (RVR)** – Rapid ventricular response
- **Ventricular Tachycardia (VT)** – Wide-complex, potentially life-threatening

Nursing Priority

- Differentiate benign sinus tachycardia from unstable arrhythmias.

CAUSES & PATHOPHYSIOLOGY

Common Causes

Physiological:
Exercise, pain,
fever,
pregnancy

Cardiac:
Ischaemia, heart
failure, valvular
disease

Non-cardiac: Anaemia,
hyperthyroidism,
hypoxia, electrolyte
imbalance

Drugs/Substances
: Beta-agonists,
caffeine, cocaine,
alcohol

Pathophysiological Mechanisms

↑ Automaticity
(sympathetic
stimulation)

Triggered activity
(electrolyte
imbalance, drug
toxicity)

Re-entry
circuits
(AVNRT, AVRT,
VT)

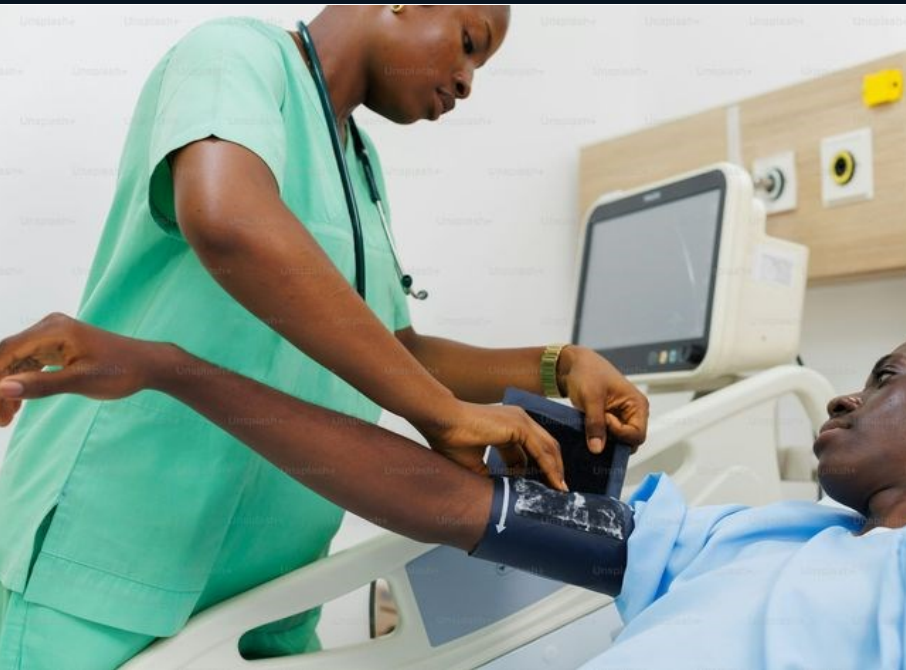
Haemodynamic Effects

↓ Diastolic
filling

↓ Coronary
perfusion

↑ Myocardial
oxygen
demand

Clinical Features & Diagnostic Investigations



- **Symptoms**
- Palpitations
- Chest pain
- Dizziness/syncope

- **Signs**
- HR >100 bpm
- Hypotension (unstable cases)
- Irregular rhythm (AF)

- **Key Investigations**

- 12-lead ECG (primary diagnostic tool)
- Continuous cardiac monitoring
- Echocardiogram
- **Nursing Role:** Early recognition of instability.





Medical Management

Initial Stabilisation (ABC)

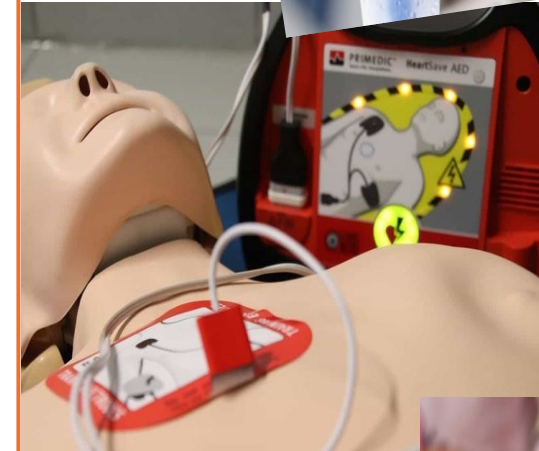
- Oxygen if needed
- IV access
- Continuous ECG monitoring

Pharmacological Treatment

- Calcium channel blockers
- Adenosine (acute SVT)
- Amiodarone (VT/AF)
- Anticoagulation (AF)

Non-Pharmacological

- Synchronized cardioversion
- Defibrillation (pulseless VT)
- Catheter ablation



Nursing Care, Prevention & Complications

Nursing Responsibilities

- Continuous monitoring of vitals & ECG
- Medication administration & safety
- Patient education & lifestyle advice

Prevention

- Manage hypertension, diabetes, obesity
- Limit caffeine, alcohol, stimulants
- Promote medication adherence

Complications

- Shock
- Heart failure
- Stroke (AF)
- Sudden cardiac death (VT)

Key Message

- Nurses play a critical role in early detection, safe management, and long-term prevention.



A stylized graphic of a stethoscope. The tubing is red and forms a large heart shape on the left side of the image. The binaural part at the top left is grey with two black earpieces. A small grey circular detail is visible on the red tubing near the bottom right.

THANK
YOU!